Simulation Time (ns)	Stimulus Event	FSM State	Counter	Output Binary (lights[5:0])	Hex Value	Active Signal Lights
0 – 50	clr = 1	S0_R1_GRN (0)	0	6'b001100	12	Road 1 **GREEN**, Road 2 **RED**
50 – 150	clr = 0, SNS2 = 0	S0_R1_GRN (0)	0 → 5	6'b001100	12	Road 1 **GREEN**, Road 2 **RED**
150 – 350	SNS2 = 1 (Car arrives)	S0_R1_GRN (0)	5 → 15	6'b001100	12	Road 1 **GREEN**, Road 2 **RED**
350 – 410	15 cycles elapsed	S1_R1_YEL (1)	0 → 3	6'b010100	20	Road 1 **YELLOW**, Road 2 **RED**
410 – 450	Yellow timer elapsed	S2_ALL_RED1 (2)	0 → 2	6'b100100	36	Road 1 **RED**, Road 2 **RED**

Simulation Time (ns)	Stimulus Event	FSM State	Counter	Output Binary (lights[5:0])	Hex Value	Active Signal Lights
450 – 600	Clearance complete, SNS2 = 0	S3_R2_GRN (3)	0 → 7	6'b100001	21	Road 1 **RED**, Road 2 **GREEN**
600 – 660	db_level1 = 95 **(Ambulance)**	S4_R2_YEL (4)	0 → 3	6'b100010	22	Road 1 **RED**, Road 2 **YELLOW**
660 – 720	Yellow clears, Siren active	S0_R1_GRN (0)	Held at 0	6'b001100	12	Road 1 **GREEN**, Road 2 **RED**
720 – 820	db_level1 = 45 (Siren ends)	S0_R1_GRN (0)	0 → 5	6'b001100	12	Road 1 **GREEN**, Road 2 **RED**
820 – 1070	SNS2 = 1 (Second car)	S0 → S1 → S2 → S3	Follows cycles	Dynamic	Dynamic	Dynamic transitions

Simulation Time (ns)	Stimulus Event	FSM State	Counter	Output Binary (lights[5:0])	Hex Value	Active Signal Lights
1270	$finish executed	—	—	—	—	—